入侵澳洲的毒蟾

The Toad That Invaded Australia



引进的灭虫帮手，为什么成了生态难题？

英文原文

Greyback cane **beetles**, whose **larvae devour** roots, were **plaguing lucrative** sugar **plantations** in Queensland, north-eastern Australia. In June 1935 Queensland **imported** 102 cane toads, **bulky amphibians native to** the Americas, from Hawaii. That August about 2,400 of their young were **released**. **Biological** control, setting one **species** against a **pest**, had an **ambitious** new **recruit**.

The **hasty** plan met **opposition**. In November 1935 an insect expert, Walter Froggatt, **persuaded** officials to **halt** releases; after Queensland's **lobbying**, they **resumed** in September 1936. The **scheme flopped**: the toads barely **dented** beetle numbers, perhaps because the insects stayed high up on cane **stalks**.

The toads **thrived** instead. With few **foes** and **fertile** females **spawning** thousands of eggs, they **multiplied unchecked**. Their **invasion frontier**, creeping about ten kilometres a year at first, **accelerated** to about 50 a year in the **tropical** north. In February 2009 the first known toad **breached** Western Australia's border.

Toxin glands in their skin make them a deadly meal. Numbers of goannas (large **lizards**) and northern quolls (small **marsupial** hunters) have **slumped** in places; some local quoll populations were **wiped out**. The toads have **wreaked havoc**, but the damage is **uneven**; some species have **prospered**.

Biological control can **triumph**. From 1926 the cactus **moth**, Cactoblastis, cleared most **prickly** pear, an **invasive cactus**, from Queensland and New South Wales within about a decade. Tested first, its **larvae starved** rather than eat fifteen other plants; the toads faced no such test. Under an October 1984 law, recruits now face **stringent scrutiny**. The **blunder** was born of **haste** and sealed by **lobbying** that **overrode** a warning: an **untested ally**, once **loose**, may never be **recalled**.

中文译文

灰背蔗甲虫它的幼虫啃食根系，正困扰着利润丰厚的甘蔗种植园，此种植园位于澳大利亚东北部的昆士兰州。1935年6月，昆士兰州从夏威夷引进了102只甘蔗蟾蜍，这是一种原产于美洲、庞大笨重的两栖动物。同年8月，它们的约2,400只后代被释放到野外。生物的防治，即用一个物种去对付一种害虫，就此迎来了一名雄心勃勃的新兵。

这个草率的计划遭到了反对。1935年11月，昆虫专家沃尔特·弗罗格特说服官员停止释放；经昆士兰州游说，释放于1936年9月恢复。这套计划/方案彻底失败：蟾蜍几乎没能削减甲虫的数量，或许是因为这些昆虫一直待在甘蔗茎秆的高处。

蟾蜍自己反倒兴旺起来。天敌寥寥，能生育的雌蟾产下数千枚卵，它们不受遏制地大量繁殖。它们的入侵前沿起初每年缓慢行进约十公里，到了热带北部，加快到每年约50公里。2009年2月，已知的第一只蟾蜍突破了西澳大利亚州的边界。

它们皮肤里的毒素腺体，使它们成为了一顿致命的餐食。巨蜥（大型蜥蜴）和北袋鼬（一种小型有袋类猎手）的数量在一些地方锐减；部分地方的袋鼬种群被彻底消灭。蟾蜍造成了严重破坏，但损害并不均衡；有些物种反而繁荣了起来。

生物防治也能大获成功。从1926年起，仙人掌螟这种蛾子在约十年内清除了大部分“带刺的梨”，这是一种外来入侵的仙人掌，清除范围遍及昆士兰州和新南威尔士州。这种蛾子放出前经过检验，其幼虫宁可饿死也不吃另外十五种植物；蟾蜍却从未受过这样的检验。根据1984年10月通过的一部法律，如今新兵都要经过严格的认真彻底的审查。这个大错起于仓促，又因游说压过了警告而成定局：未经检验的盟友一旦放出，就可能再也召不回来。

词汇与表达

beetles n.
甲虫

larvae n.
幼虫
whose larvae devour roots

devour v.
啃食
larvae devour roots

plaguing v.
困扰

lucrative adj.
利润丰厚的
lucrative sugar plantations

plantations n.
种植园

imported v.
引进了

bulky adj.
庞大笨重的

amphibians n.
两栖动物
bulky amphibians

native to phr.
原产于
native to the Americas

released v.
释放

Biological adj.
生物的

species n.
物种
setting one species against a pest

pest n.
害虫
one species against a pest

ambitious adj.
雄心勃勃的

recruit n.
新兵

hasty adj.
草率的
The hasty plan met opposition

opposition n.
反对
The hasty plan met opposition

persuaded v.
说服
persuaded officials to halt releases

halt v.
停止

词汇与表达（续）

lobbying n.
游说
after Queensland's lobbying

resumed v.
恢复
they resumed in September 1936

scheme n.
计划/方案

flopped v.
彻底失败

dented v.
削减

stalks n.
茎秆

thrived v.
兴旺
The toads thrived instead

foes n.
天敌

fertile adj.
能生育的

spawning v.
产下

multiplied v.
大量繁殖

unchecked adj.
不受遏制地
they multiplied unchecked

invasion n.
入侵
Their invasion frontier

frontier n.
前沿

accelerated v.
加快
accelerated to about 50 a year

tropical adj.
热带

breached v.
突破了

Toxin n.
毒素
Toxin glands in their skin

glands n.
腺体

lizards n.
蜥蜴

marsupial adj.
有袋类

词汇与表达（续）

slumped v.
锐减

wiped out phr.
彻底消灭
quoll populations were wiped out

wreaked havoc phr.
造成了……严重破坏
The toads have wreaked havoc, but the damage is uneven;

uneven adj.
不均衡

prospered v.
繁荣了

triumph v.
大获成功

cactus n.
仙人掌

moth n.
蛾子

prickly adj.
带刺的

invasive adj.
外来入侵的
an invasive cactus

starved v.
饿死

stringent adj.
严格的
stringent scrutiny

scrutiny n.
认真彻底的审查
stringent scrutiny

blunder n.
大错

haste n.
仓促

overrode v.
压过了
lobbying that overrode a warning

untested adj.
未经检验的

ally n.
盟友

loose adj.
放出

recalled v.
召不回来

文章简述

- 引进：1935年，澳大利亚为对付危害甘蔗的甲虫，引进甘蔗蟾蜍，再把约2400只后代放进野外，希望它们吃掉害虫。
- 冒进：专家曾促成暂停释放，但地方游说让计划恢复；放出前却缺少检查蟾蜍究竟吃什么的试验。
- 失控：甲虫没被明显消灭，蟾蜍却因天敌少、繁殖快而不断扩散，在热带北部每年推进约50公里。
- 伤害：蟾蜍皮肤有毒，捕食它们的巨蜥、北袋鼬在一些地方大幅减少，部分袋鼬种群消失；对不同物种的影响并不一致。
- 教训：此前引进仙人掌螟时，试验发现它的幼虫宁可饿死也不吃另外15种植物，后来才用于控制仙人掌；蟾蜍没有经过同类食性检验。未经检验的“帮手”一旦失控，可能再也收不回来。